

GroAurum

Phase 2.5 Completion Summary

Database proof: integration tests, RLS validation, local Supabase workflow

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Live proof commit: ca2967b

Phase 2.5 proves the Supabase/PostgreSQL schema with live integration tests and local tooling. Customer screens still use the legacy mock adapter. Phase 3 not started.

1. Phase 2.5 Goal

Validate the Phase 2 schema against a real local Supabase/PostgreSQL environment: apply migrations, generate database types, test business invariants, and prove RLS role boundaries. Validation and correction only - no customer Supabase wiring.

2. What Was Added

- @groaurum/db-tests - live PostgreSQL + Supabase Auth integration test package
- Root scripts: db:start, db:stop, db:reset, db:status, db:types, test:db, db:proof
- scripts/check-db-prerequisites.mjs - Docker/local Supabase/production URL guards
- scripts/generate-db-types.mjs - CLI-generated database.generated.ts workflow
- Migration 014 - trusted sku_prices row closure (append-only price history fix)
- Migration 015 - API role grants for PostgREST authenticated/anon clients
- docs/architecture/phase-2.5-database-proof.md
- .env.test.example - safe local test configuration template
- supabase as project devDependency (pnpm exec supabase)

3. Local Tooling Status

Tool	Status
Node.js	v24.15.0 - available
pnpm	9.15.4 - available
Supabase CLI	Available via pnpm exec supabase
Docker Desktop	Reachable - Engine 29.6.1 (local Supabase verified)

Live database proof completed: pnpm db:start, pnpm db:reset, pnpm db:types, and pnpm test:db (34/34) all passed against local PostgreSQL.

4. Local Supabase Commands

```
pnpm db:start      # start local stack
pnpm db:reset      # clean replay migrations 000-015
pnpm db:status     # local URLs and keys
pnpm db:types      # generate database.generated.ts
pnpm test:db       # integration + RLS tests
pnpm db:proof      # reset + types + tests
```

5. Migration Corrections

014 - sku_prices trusted row closure:

- Blocked: mutating trade_price, sku_id, or other commercial fields on sku_prices
- Allowed (trusted only): setting effective_to to close the open row before new insert
- Replaces blanket prevent_modification UPDATE trigger on sku_prices

015 - API role grants:

- Grants authenticated/anon table privileges required for PostgREST clients
- Fixes permission denied for table despite RLS policies being present
- RLS still enforces row scope; grants only enable client access to tables

6. Database Invariant Tests

- Empty catalogue - zero categories/products/skus/prices is valid
- DELIVERED requires PAID - UNPAID rejected; PAID allowed via trusted workflow
- Order snapshot immutability - commercial fields locked after CONFIRMED
- Price history append-only - UPDATE/DELETE on trade fields rejected; new row for new price
- Inventory movement append-only - UPDATE and DELETE rejected
- Audit log append-only - UPDATE and DELETE rejected
- Zero credit - no CREDIT enum; invalid payment intent insert fails

7. RLS Behaviour Tests

Tests use real authenticated Supabase clients - not SQL text presence only.

- CUSTOMER: read linked shop/orders; cannot read other customers; no price/order status writes
- SALESMAN: assigned shop access only; cannot mark DELIVERED; no payment completion writes
- DELIVERY: assigned route access only; cannot change prices or order totals; no DELIVERED bypass
- ADMIN: catalogue writes allowed; append-only protections still apply on sku_prices

8. Mobile Normalization and Shop Linking

Canonical stored format: E.164 +91XXXXXXXXXX

Input	Canonical
9876543210	+919876543210
+91 98765 43210	+919876543210
919876543210	+919876543210
09876543210	+919876543210

shop_auth_links enforces one active shop per auth user (unique on auth_user_id). OTP activation service not implemented - database proof only.

9. Service Area Schema Proof

- Active and inactive service_areas accepted
- PIN_CODE rules require config.pinCodes array with entries
- ADMIN_AREA rules require config.areaCodes array with entries
- Invalid/empty rule configs rejected by CHECK constraints
- Test fixtures only - no production Gurugram seeds committed

10. Trusted Workflow Test Context

```
SET LOCAL groaurum.trusted_server_action = 'true';
-- order status, payment completion, sku_prices effective_to closure
```

Tests use withTrusted() helper - same session flag intended for SECURITY DEFINER RPCs.

11. Generated DB Type Status

database.generated.ts is CLI-generated and committed after successful pnpm db:types against the live local schema. No handwritten types labelled as generated.

```
pnpm db:start
pnpm db:reset
pnpm db:types
```

12. Validation Results

Command	Result
pnpm typecheck	Pass (all workspaces)
pnpm lint	Pass
pnpm test	Pass - 14 domain + 9 schema file tests
pnpm test:db	Pass - 34/34 live invariant + RLS tests
pnpm db:reset	Pass - migrations 000-015 clean replay
pnpm --filter @groaurum/customer build	Pass - Expo web export, 14 routes

13. Safety Guards

- GROAURUM_DB_TEST_ALLOWED=true required for database tests
- SUPABASE_DB_URL must target 127.0.0.1:54322 (local only)
- Remote supabase.co hosts rejected by prerequisite script
- No production credentials committed

14. What Phase 2.5 Did NOT Do

- No Supabase wiring in customer screens
- No mock adapter replacement (Phase 3)
- No Admin, Sales, or Delivery feature UI
- No payment gateway or SMS integration
- No hard-coded catalogue products
- Phase 3 not started

15. Git History

```
18ee50a checkpoint: preserve original customer MVP
eb05c74 feat: add GroAurum B2B domain and service boundaries
fbcddb7 feat: scaffold monorepo and Supabase B2B schema
726e175 test: prove Supabase schema invariants and RLS
ca2967b test: complete live Supabase database proof
```

16. Remaining Before Phase 3

- Implement Supabase-backed service adapters in @groaurum/api-client
- Replace mock adapter behind feature flag (customer screens)
- Wire shop auth and service area gating in customer flows
- OTP activation service (shop invitation to auth link)

17. Next Step (Phase 3 - not started)

Replace not-implemented adapters with Supabase-backed implementations. Gate customer flows on shop auth and service area. Customer screens still use legacy mock until Phase 3 cutover.